

# Helping Hand Beta Build Report

M4 milestone status and evidence | Branch gael | Snapshot 02 October 2026

Helping Hand combines a wearable sign-learning glove with an Android Flutter application. The beta goal is dynamic word recognition from real sign sequences using flex and IMU data, alongside a refined learner interface and account flow. The static letter/number classifier remains the regression baseline. The current repository snapshot is ready for structured beta evaluation, but dynamic word recognition and physical-device acceptance are not yet demonstrated.

## INTEGRATED LEARNER EXPERIENCE

The app opens to email/password sign-in or account creation, with password reset and sign-out. Home, Alphabet, Numbers, and Words use a shared ivory/teal/mint visual system. Alphabet and Numbers present target grids; Words presents eight provisional vocabulary rows. Selecting any of the 44 targets opens a dedicated practice route with a title and Back control. BLE state remains owned by the app shell and refreshes the visible lesson through shared practice state.

Letter and number practice retains the static classifier feedback and completion gate: at least 80% confidence, five matching packets, and a 750 ms stable hold. Progress is stored locally under the signed-in Firebase UID and synchronized through the existing Firestore progress path. The practice page does not create a second BLE connection.

## EVIDENCE RECORDED

| Evidence                | Recorded result and limit                                                                                                                                                                                                                         |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9c5caab visual redesign | Flutter analysis, production web build, and five temporary responsive/interaction review checks passed. Checks covered phone, landscape, tablet, 3x text, reduced motion, selection/Retry callbacks, and idle/complete presentation.              |
| d581d19 route update    | Targeted Flutter analysis passed. This is source-analysis evidence; physical navigation, BLE-on-route, and Android beta validation were not run.                                                                                                  |
| M3 alpha baseline       | README records Sept. 10, 2026 checks: Flutter analyze; 16 Flutter tests; debug APK; ESP32-S3 build; 23 backend tests. It also records a historical physical BLE/static prediction demonstration. These are alpha results, not fresh beta results. |
| Static-sign pilot       | Repository export dated Sept. 17, 2026: 35 trials, 5,518 valid BLE frames, one signer, neutral orientation, about 39.7-40.5 Hz; five flex and six IMU channels are present. This is recording evidence, not word-model evidence.                  |
| Account work            | Team notes report email/password account creation and sign-in on Oct. 1. Eight UID-isolation repository tests passed at 873b589. Device/cold-restart and live Firebase cross-account behavior remain unverified.                                  |

## SCOPE BOUNDARY

Dynamic word recognition is a beta objective. In this snapshot, word pages are UI-only: no trained/validated model, word-completion storage, approved vocabulary, or target-specific ASL instruction is connected.

# Beta gates and known limitations

M4 criteria: usability, responsiveness, robustness, integrated features, and persistent state

## WHAT THIS BUILD PROVIDES

- Predictable navigation from Home and bottom tabs to target pickers and separate practice pages; Back returns to the originating picker.
- Visible connection, prediction, confidence, hold, Retry, and completion feedback for static letters and numbers.
- UID-scoped local progress plus the existing Firestore progress document; account creation, sign-in, reset, and sign-out flows are present.
- Asynchronous BLE and progress updates remain connected to the learner interface. The static 36-class model remains the regression baseline.

## REQUIRED BETA VALIDATION AND KNOWN ISSUES

| Gate                        | Current status / beta action                                                                                                                                                                                                                                   |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Current routes and controls | NAV-01 to NAV-03, LIVE-01 to LIVE-03, WORD-01, UI-01 are Not run on a physical device. Verify system Back, scroll restoration, BLE refresh, Retry, accessibility, and 44/44 target routes.                                                                     |
| Completion lifecycle        | Static target selection persists after Back. Matching packets can still complete the selected target after leaving its lesson. Team decision and user expectation remain open.                                                                                 |
| Account and cloud state     | Email/password success is team-reported; exact build/device and cold-restart evidence are absent. Firebase cross-account isolation, offline recovery, live sync, and current rules need end-to-end beta runs.                                                  |
| Hardware and release        | No fresh Android build or physical glove regression evidence for the route update. IMU part/magnetometer capability and standalone boot must be checked. Current README also records debug signing and missing packet checksum.                                |
| Usability and content       | External-user tests, on-device accessibility, and target-specific reviewed ASL instruction are outstanding. The provisional eight-word list needs ASL review.                                                                                                  |
| Dynamic words (beta goal)   | Real sequence capture with IMU data, model comparison, deployment, and word completion are beta objectives. The current pilot is static-sign data; no trained/validated/deployed word model is connected. Complete ML-01 to ML-09 before claiming recognition. |

## TRACEABILITY AND EFFORT EVIDENCE

The separate practice route and account-scoped progress connect the external interface with BLE processing and persistent state; current end-to-end behavior still needs device and cloud validation. The static learner loop is integrated. Word-level recognition is in the beta target scope but remains unimplemented in the current snapshot. Timestamped repository history provides evidence of changes (including commits 9c5caab and d581d19); the team must add person-hour records, test artifacts, known-issue updates, and stakeholder defense details before submission.

Repository: <https://github.com/GG1627/helping-hand> | Current work: branch gael, commit 14f7aad. The two PDF reports summarize the snapshot; run evidence and configuration should be appended as beta tests execute.